

Database Schema, ERD & Class Diagram

Project: Enterprise IT Support Ticketing System / Phase 1 Deliverable

Entities

users

Column	Type	Constraints	Notes
id	INTEGER	PK, AUTOINCREMENT	
name	TEXT	NOT NULL	
email	TEXT	UNIQUE, NOT NULL	Login identifier
password_hash	TEXT	NOT NULL	bcrypt hash
role	TEXT	NOT NULL, CHECK IN ('employee','agent')	
created_at	TEXT	NOT NULL, default now()	ISO-8601 string

tickets

Column	Type	Constraints	Notes
id	INTEGER	PK, AUTOINCREMENT	
title	TEXT	NOT NULL	
description	TEXT	NOT NULL	
priority	TEXT	NOT NULL, CHECK	Low Medium High Critical
category	TEXT	NOT NULL, CHECK	Hardware Software Network Access
status	TEXT	NOT NULL, default 'open'	open in_progress resolved closed
created_by	INTEGER	FK → users.id, NOT NULL	Employee who filed the ticket
assignee_id	INTEGER	FK → users.id, NULLABLE	Unassigned by default
created_at	TEXT	NOT NULL, default now()	
updated_at	TEXT	NOT NULL, default now()	Updated on status change

comments

Column	Type	Constraints	Notes
id	INTEGER	PK, AUTOINCREMENT	
ticket_id	INTEGER	FK → tickets.id, NOT NULL	Cascade delete with ticket
user_id	INTEGER	FK → users.id, NOT NULL	Author (employee or agent)
body	TEXT	NOT NULL	
created_at	TEXT	NOT NULL, default now()	Orders the thread

attachments (stretch goal)

Column	Type	Constraints	Notes
id	INTEGER	PK, AUTOINCREMENT	
ticket_id	INTEGER	FK → tickets.id, NOT NULL	
file_url	TEXT	NOT NULL	Path under /uploads served by API
uploaded_at	TEXT	NOT NULL, default now()	

Relationships (ERD)

- users (1) —< tickets — via tickets.created_by (one employee files many tickets)
- users (1) —< tickets — via tickets.assignee_id (one agent is assigned many tickets, nullable)
- tickets (1) —< comments — via comments.ticket_id (one ticket has many comments)
- users (1) —< comments — via comments.user_id (one user authors many comments)
- tickets (1) —< attachments — via attachments.ticket_id (one ticket has many attachments)

Class Diagram

